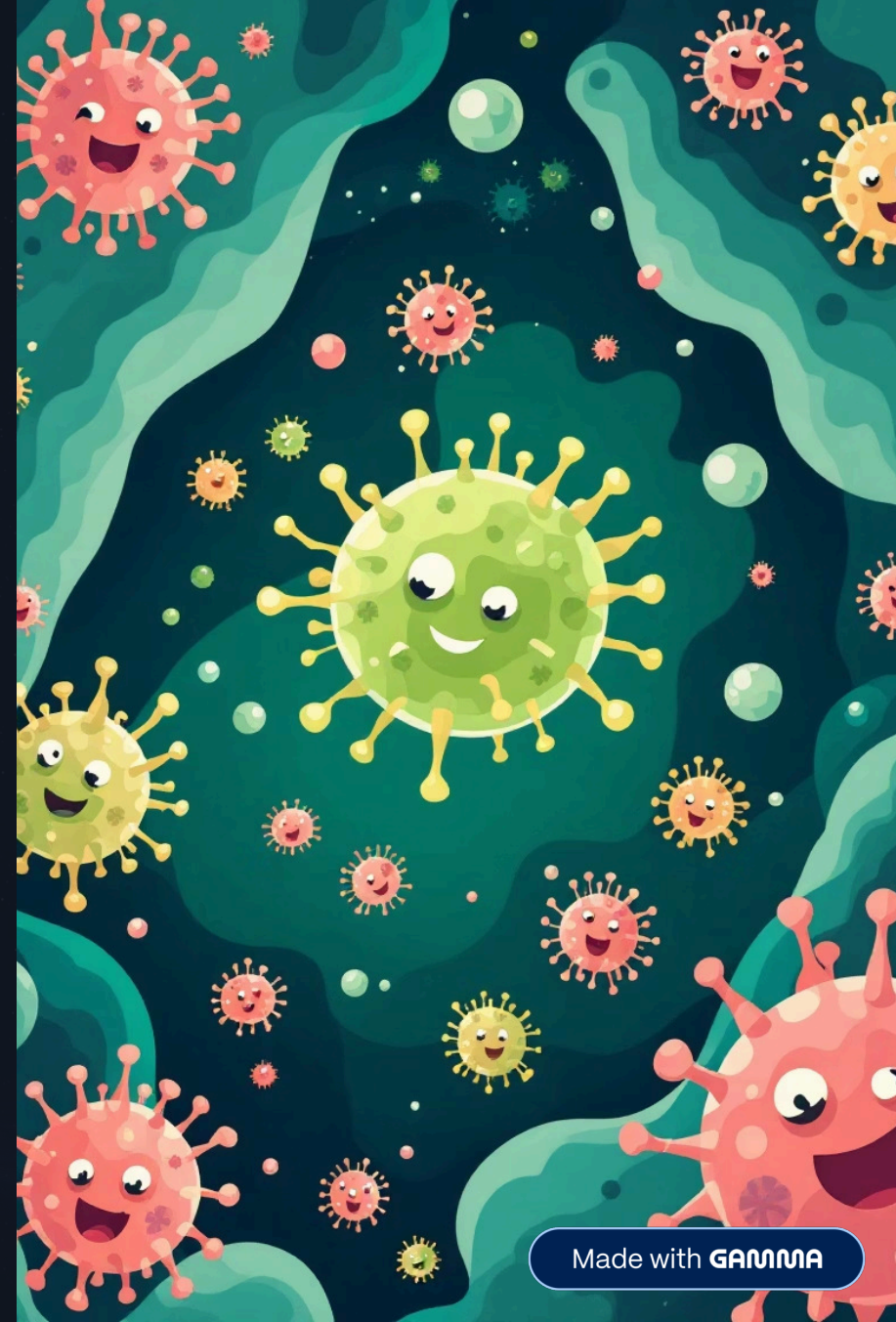


# Seasonal Flu Vaccine Prediction

Data-driven insights to improve vaccination outreach and public health outcomes

PHASE 3 ANALYSIS



Made with GAMMA



DATA FOUNDATION

# National 2009 H1N1 Flu Survey

## Dataset Scale

- 26,707 survey respondents
- 35 features analyzed
- Demographics, health status, beliefs
- Binary classification target

## Feature Categories

- Demographics & health status
- Medical history & behaviors
- Vaccine opinions & beliefs
- Risk factors & social contact

# Understanding Vaccination Behavior

## Challenge

Limited resources for vaccine promotion campaigns

## Solution

Predict who won't get vaccinated to target outreach effectively

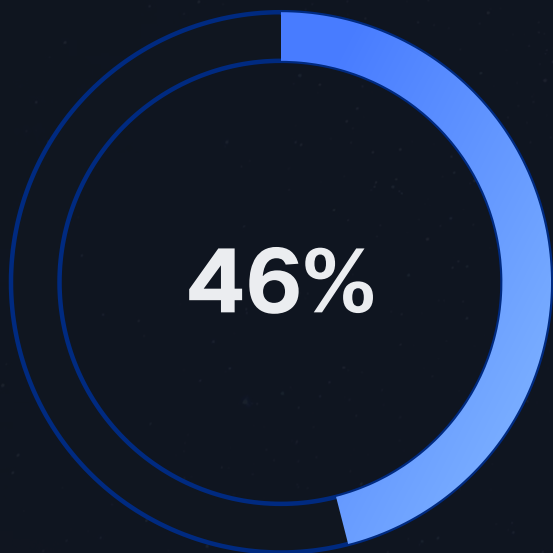
## Impact

Improve vaccine adoption during pandemics



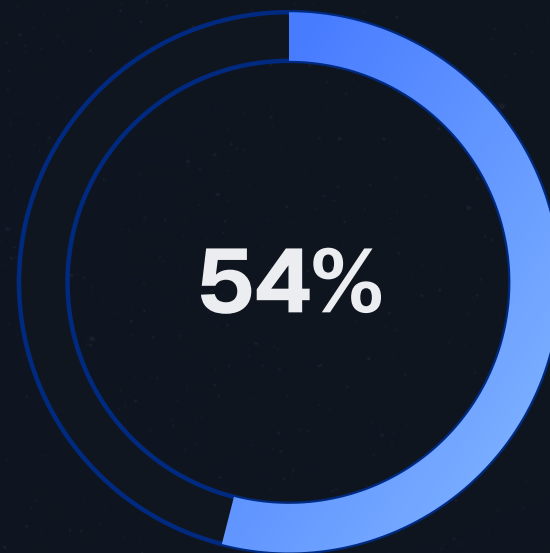


# Vaccination Distribution



**Vaccinated**

Received seasonal flu vaccine

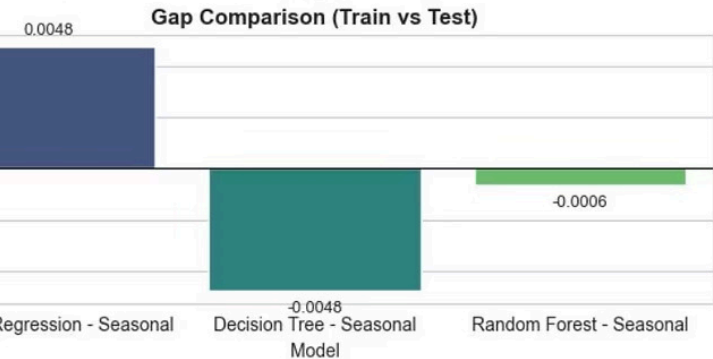
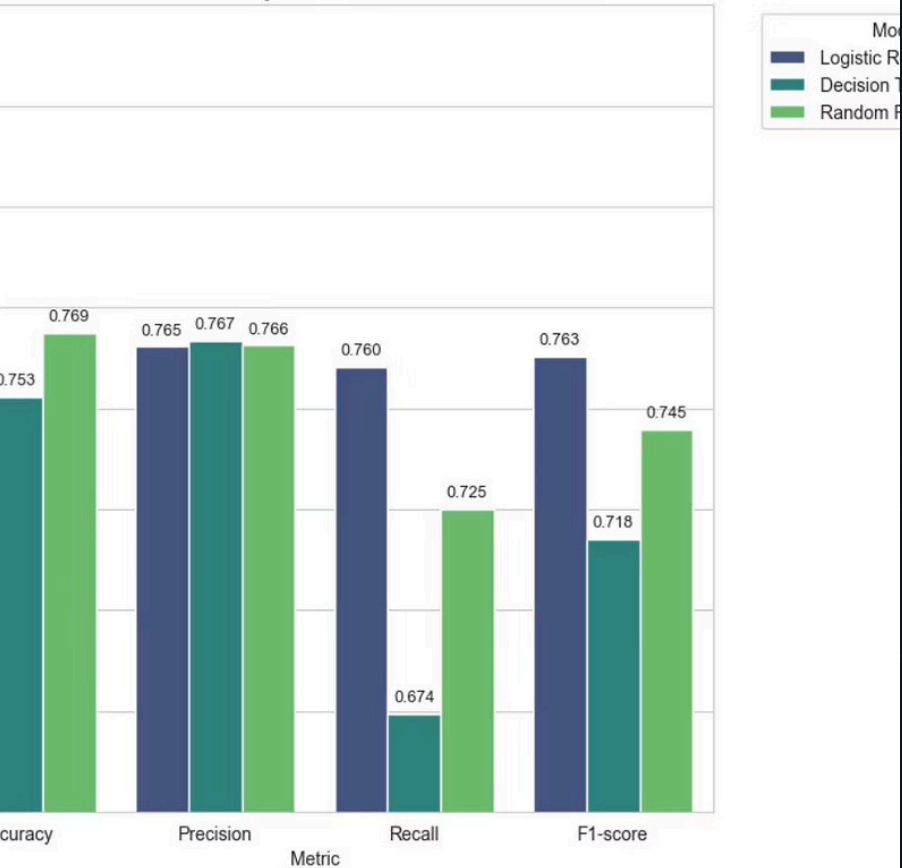


**Not Vaccinated**

Did not receive vaccine

Significant opportunity to improve vaccination rates through targeted interventions

Performance Comparison: Seasonal Models



# Model Performance Comparison

78%

**Logistic Regression**  
Best overall accuracy

76%

**Recall Score**  
Identifies vaccine recipients

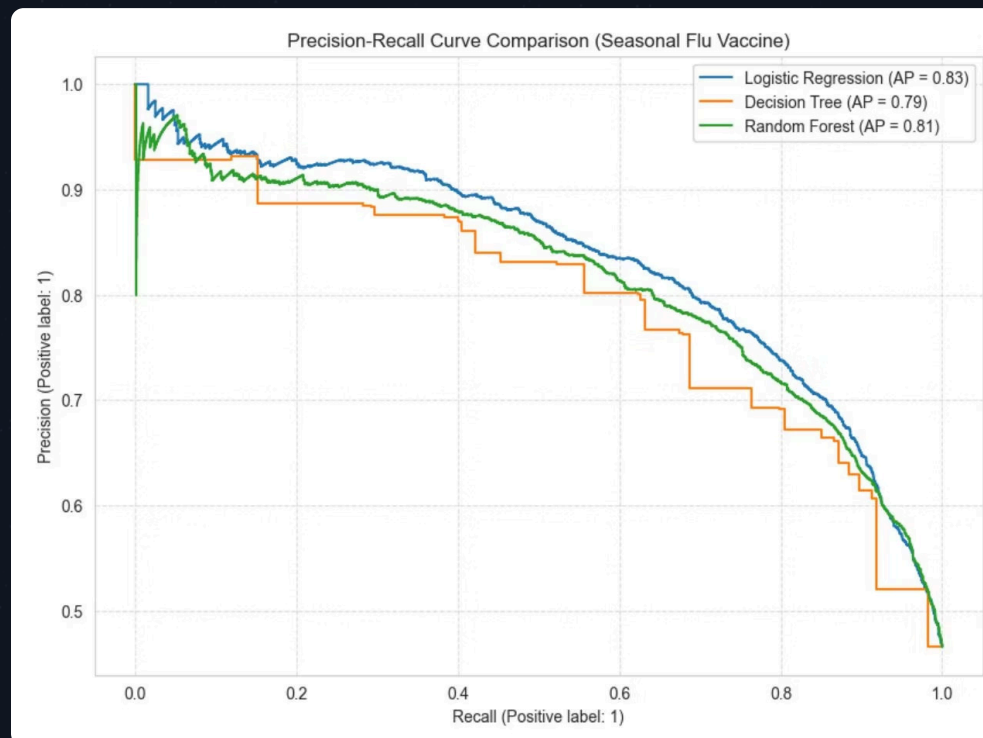
76%

**F1-Score**  
Balanced performance

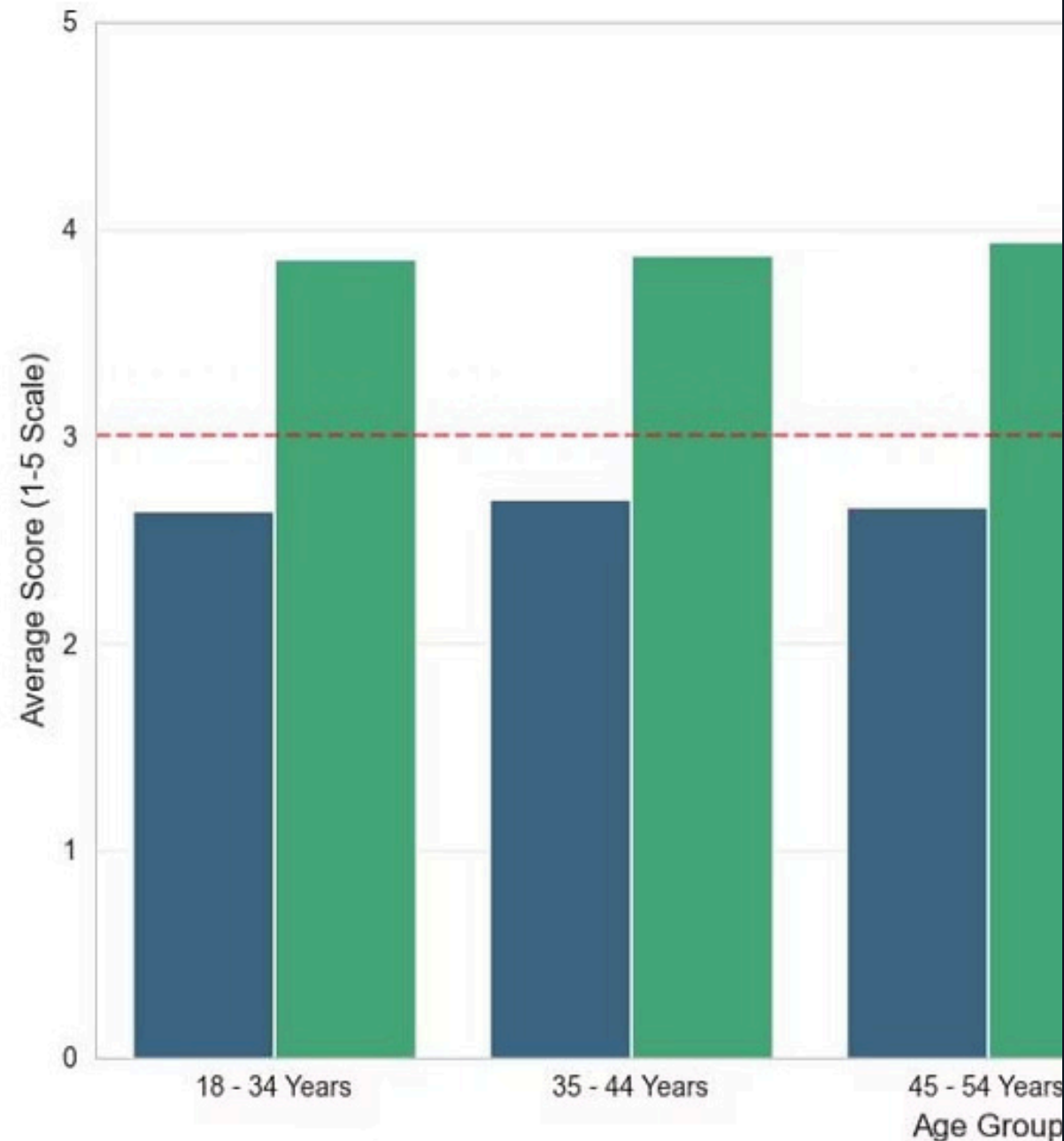
# Logistic Regression: The Clear Choice

## Why This Model Wins

- Highest accuracy at 78%
- Balanced precision & recall at 76%
- Minimal overfitting (0.5% gap)
- Identifies 76% of vaccine recipients
- Strong interpretability for stakeholders



- Average Precision of 0.76 indicates excellent balance between catching all vaccine recipients and maintaining prediction confidence



# Public Sentiment Analysis

1

## 18-34 Years

Lowest engagement & vaccination rates

2

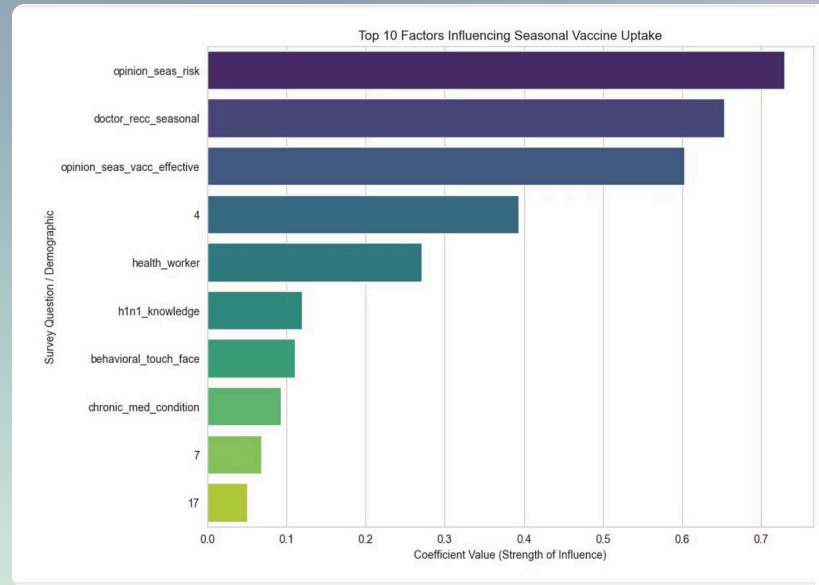
## Critical Gap

Low risk perception despite high vulnerability

3

## Trust Increases

Vaccine confidence rises with age



## INSIGHTS

# What Drives Vaccination?

## Top Influencers

Doctor recommendations and perceived vaccine effectiveness are the strongest predictors of vaccination uptake

## Key Insight

Trust in medical professionals and belief in vaccine efficacy drive behavior more than demographics





## ACTION PLAN

# Strategic Recommendations

01

## Target Low-Uptake Groups

Focus on 18-34 age demographic with lowest engagement rates

02

## Launch Severity Campaigns

Highlight flu hospitalization statistics and serious health risks

03

## Leverage Social Media

Deploy digital campaigns to reach younger demographics effectively

04

## Shift Messaging Focus

Move from "where to get vaccine" to "why flu is serious risk"

# Impact & Next Steps



## Precision Targeting

76% recall enables identification of most at-risk individuals for intervention



## Public Health Gains

Data-driven outreach improves vaccination rates and pandemic preparedness



## Scalable Solution

Model framework applicable to future vaccination campaigns

Empowering public health organizations with predictive insights to save lives through targeted vaccine outreach

